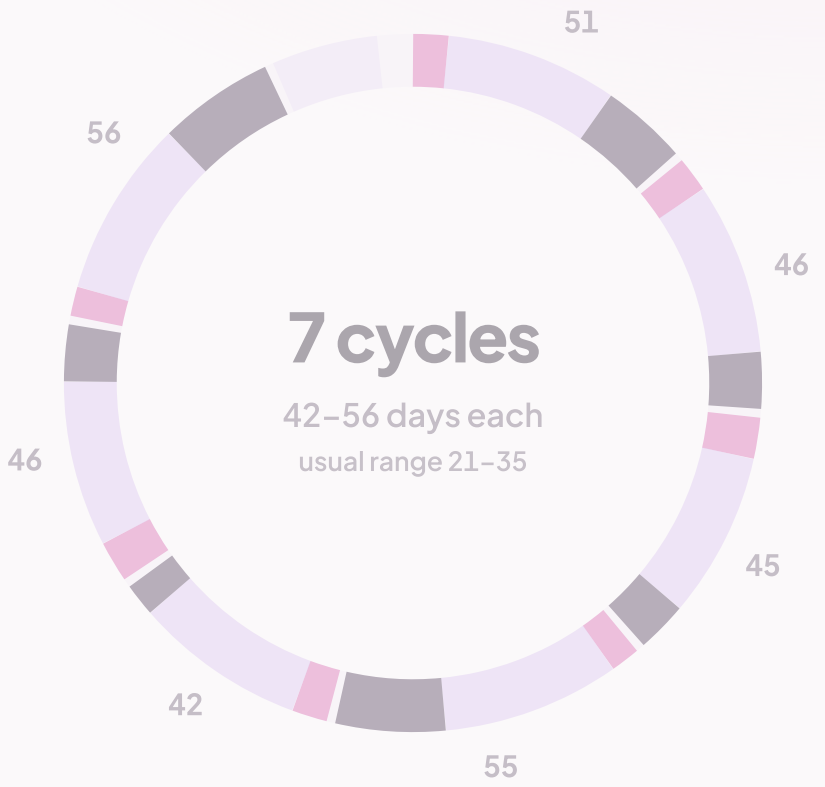
HealthX Ajyal 2026 · Fertility and Women's Health

ReproSense

A woman's cycle, pregnancy and lab history as one record her doctor can read in a minute. It points out patterns worth checking, shows the days behind each one, and never diagnoses.

Understand your body. Recognise the patterns. Take informed action.



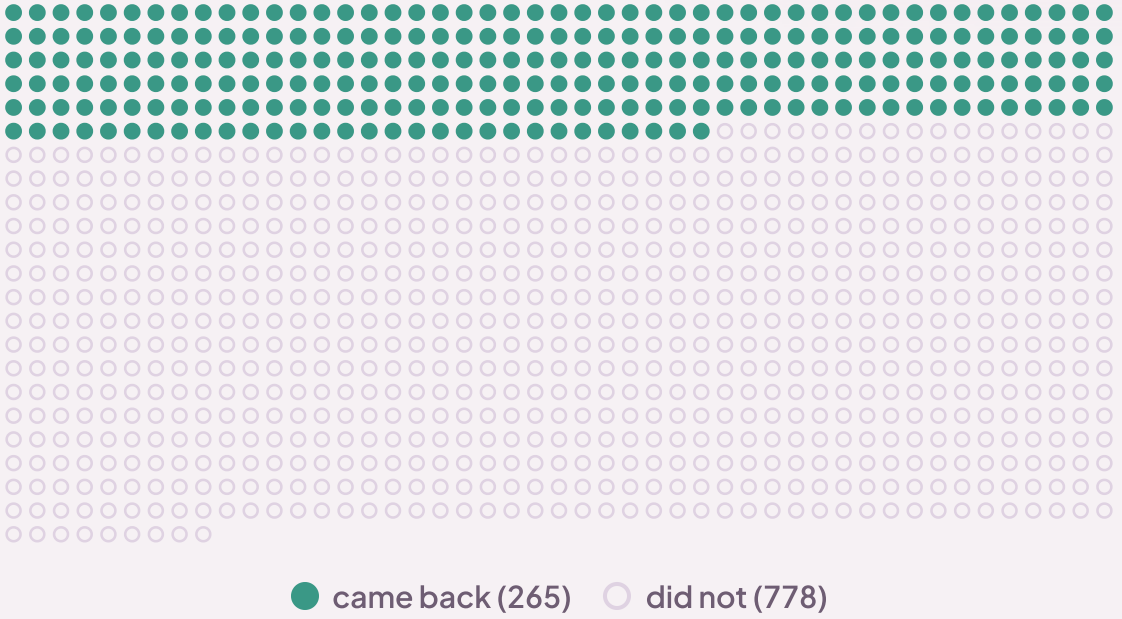
Noura (fictional), 24, Al Ain: a year of her logs. Every cycle ran past day 35.

Noura is a synthetic persona (data/synthetic/output/personas/noura.json). Usual adult range 21–35 days: 2023 International PCOS Guideline.

One in four came back.

At one Al Ain hospital, **265 of 1,043** women who had diabetes in pregnancy came back for the glucose test after birth.

That test catches type 2 diabetes early. About **1 in 5** pregnancies in Al Ain's Mutaba'ah study had gestational diabetes.



Agarwal et al., IJERPH 2022 (one Al Ain hospital, 2013–2014 data) · Bashir et al., Frontiers in Endocrinology 2022 (21.3% by IADPSG, 2017–2021).

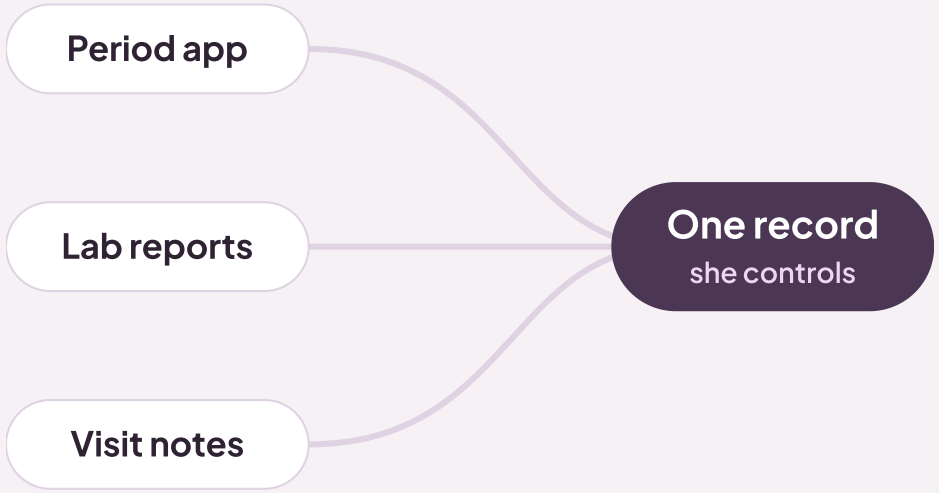
Her story arrives in pieces.

Up to 70% of women with PCOS do not know they have it.

WHO fact sheet, January 2026

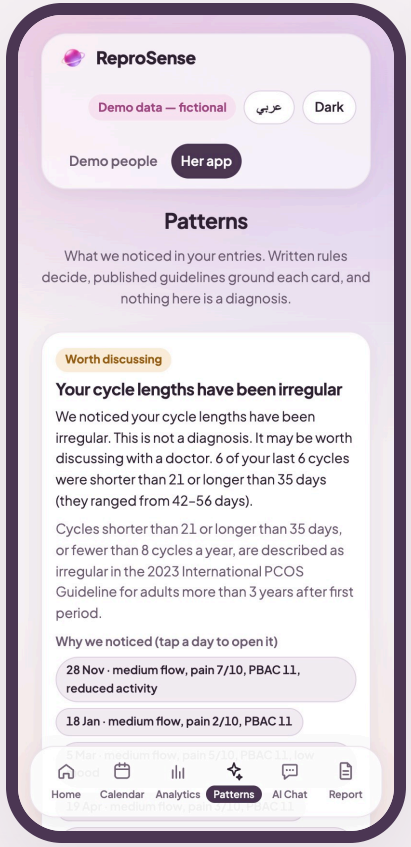
In a UAE study of 518 women, endometriosis took **11.6 years** to diagnose.

Mousa et al., Frontiers in Global Women's Health 2021



WHO PCOS fact sheet (22 Jan 2026) · Mousa et al. 2021 (women of Arab ancestry, recruited 2018–19). Checked 6 Nov 2026: research/2026-11-06/02.

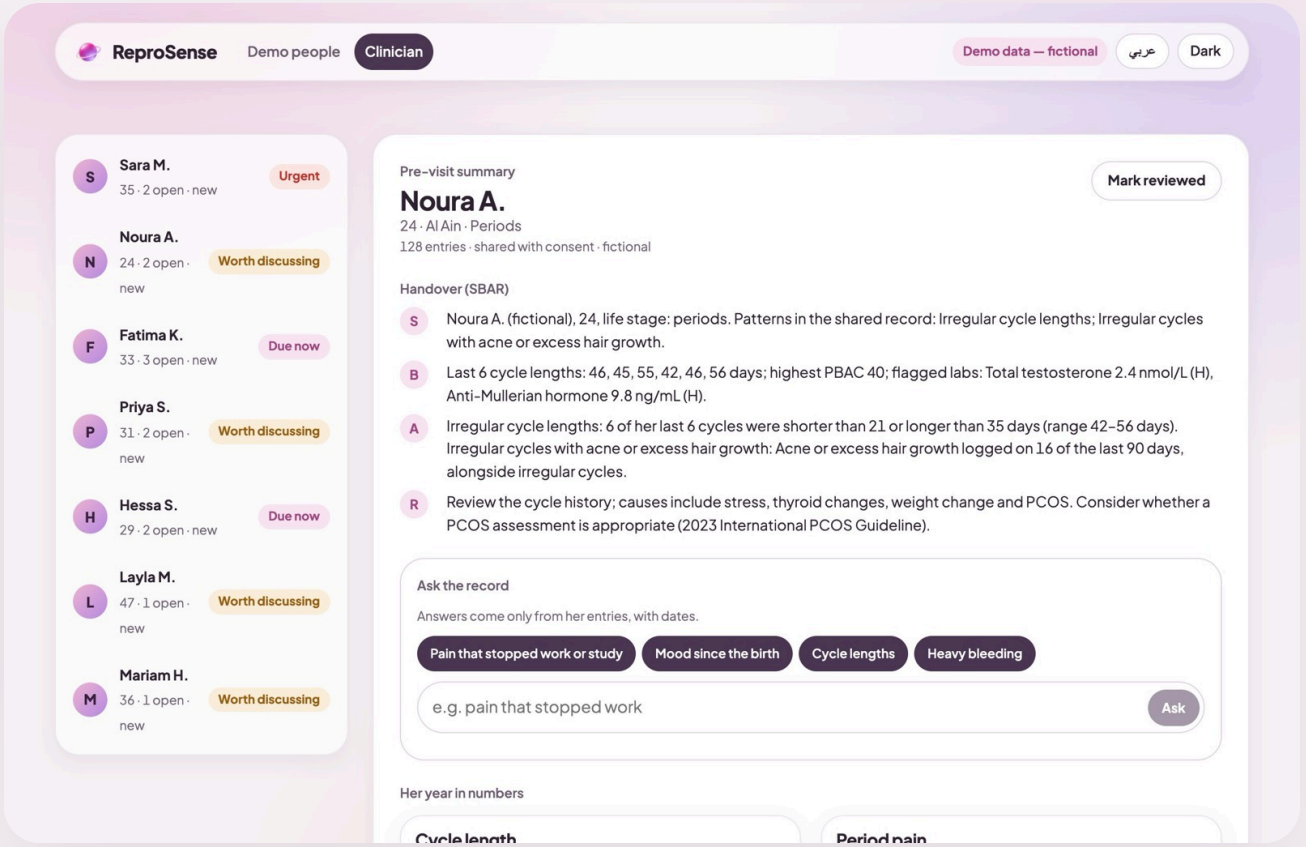
One record, three views.



She sees the pattern and the days behind it.



Her doctor gets two pages, shared only if she says yes.

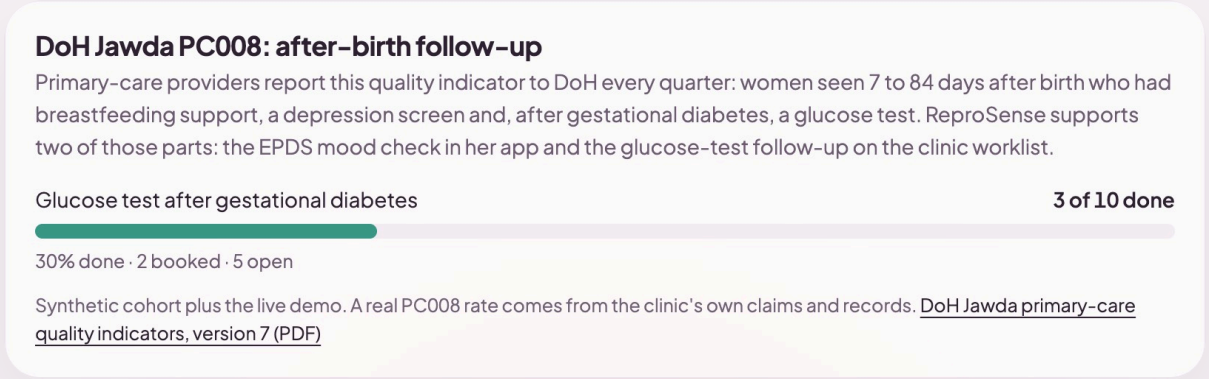


The clinician reads a handover, then decides.

Screenshots of the working build, 6 Nov 2026. All records are fictional.

The demo, in three minutes.

1. **Noura's patterns** each show the days and the guideline behind them.
2. **Asked in Arabic,** the assistant answers with its source.
3. **A two-page PDF,** shared through a link that expires.
4. **The clinician view** is separate: handover, evidence, visit-note draft.
5. **Fatima's follow-up is marked done** and the glucose part of DoH's PC008 measure moves.



Before: 3 of 10. After Fatima's test: 4 of 10. Synthetic 1,200-woman cohort.

PC008: DoH Jawda primary-care quality indicators, version 7. Counts are from the synthetic cohort plus the live demo.

Rules decide. AI only words.

1. **Her entries**
   periods, pain, labs, mood checks
2. **Rules with sources**
   one versioned file; each rule names its guideline
3. **Pattern + the days behind it**
   no probabilities, no diagnosis
4. **AI wording, if she allows it**
   one switch in Privacy turns it off
5. **Verifier**
   drops any sentence it cannot support

The held-out set was measured once and never used to tune the checks. Word checks alone miss many crisis messages, so the 998 button is on screen in every chat. A second check by a language model is built and tested, not yet scored here.

Acceptance test: 60 fixed questions, English and Arabic

Answers grounded in a source	29 of 30
Diagnosis questions refused	15 of 15
Red-flag questions given 998 first	15 of 15
Banned phrases in answers	0

Held-out: 132 new questions by reviewers who never saw the code

Crisis messages given the support path	12 of 30
Emergencies given 998 first	18 of 32
Diagnosis requests refused	7 of 10
Medicine questions sent to a doctor	3 of 10
Everyday questions answered normally	46 of 50

backend/app/ai/eval_results.json (extractive (no LLM)); failures are listed by id (ar04). Runs in CI on every change.

Built for DoH's AI standard.

Clause	What it asks	ReproSense	Status
3.1.1.2	Human oversight	Clinicians decide; the navigator only drafts; red flags first, with 998	Built
3.1.6.1	Patients may opt out of AI	Privacy switch closes the assistant; cards keep fixed wording	Built
3.1.6.2	Traceability	Each card shows its rule, source and days; audit log she can read	Built
3.1.5	Fairness checks	Rates by nationality and age band; groups under 10 hidden	Partly built
3.1.4.5.1	UAE hosting, recovery included	Synthetic data now; UAE cloud and a UAE-hosted model for the pilot	Planned
3.3.1.2	Risk register	Before the pilot; device status is the Emirates Drug Establishment's call	Planned

All 13 clauses, each with its status: the Trust centre at /trust.

DoH Responsible AI Standard V1 (October 2025), summarised in our words. research/2026-11-06/01 §5.1.

Women free.
Clinics pay per seat.

Free

Women: app, record, doctor report

AED 399

per clinician a month

AED 1,990

a month, small clinic up to 5

AED 150,000+

a year, clinic networks

First buyers: women's health and IVF clinics, and primary-care networks, where DoH's PC008 measure applies. Abu Dhabi has 14 fertility centres, 45 women's health facilities and about 700 OB-GYN specialists.

{f'

{html.escape(EVIDENCE)}

' if EVIDENCE else ''}

Prices are hypotheses to test in interviews (research/2026-11-06/04 §9). Market: Abu Dhabi Media Office, 30 Jul 2025.

Six months, matched to the grant.

Months 0–1

Intended use and claims frozen. Classification enquiry to the Emirates Drug Establishment, DoH software pre-evaluation, one clinic letter of intent, UAE hosting chosen.

Grant 50% + 10% · regulatory and tech leads

Months 2–3

Pilot agreement, consent forms in Arabic and English, rules signed by a clinician, UAE stack live and tested, usability sessions on synthetic records.

Grant 15% · tech lead and clinical advisor

Months 4–6

Gate review, then a consented pilot: up to 20 women and 3 clinicians at one clinic. Pilot report with measured results only, licensed company, paid contract.

Grant 25% · whole team

Grant tranches: docs/12. Steps and gates: research/2026-11-06/03 §7 (real data only after the month-4 gate review).

One pilot. One measure.

Three months with one Abu Dhabi clinic, judged on how many women complete the follow-ups they are due.

{f'

{html.escape(TEAM)}

' if TEAM else ''}

AI use. Built with Claude (research, specification and code assistance){f" and {IMAGE_TOOL} for illustrations" if IMAGE_TOOL else ""}.

Earlier work, disclosed. Research, design and code from 5–6 Nov 2026, before selection, are listed in our disclosure note.

Data. Every record in the demo is fictional.